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Curiosa mathematica, Part 1 : A new theory of parallels

Lewis Carroll

1895

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A NEW THEORY

OF

PARALLELS

In every Circle, the inscribed equilateral Tetragon

is greater than any one of the Segments which lie outside it.

Curiosa Mathematica

PART I

A NEW THEORY

OF

PARALLELS

BY

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FOURTH EDITION

PRICE TWO SHILLINGS

London

MACMILLAN AND CO.

1895

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Oxford

HORACE HART, PRINTER TO THE UNIVERSITY

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PREFACE TO THIRD EDITION.

The chief novelty, in the First Edition of this treatise, was the Axiom, by means of which I proved Euc. I. 32 without making use of his 12th Axiom. And the chief novelty, in this Third Edition, is the change I have made in that Axiom, by substituting 'Tetragon' for 'Hexagon'. The new Figure is more simple, and more easily constructed, than its predecessor: while the Axiom is, I hope, as obviously true as ever.

The proof of my "New Theory of Parallels" is, I think, greatly simplified and improved in this new Edition—the Propositions, which do not require any disputable Axiom, being placed by themselves in 'Book I,' while those, which require the new Axiom for their proof, are placed in 'Book II.' At the end of Book II will be found a proof (so far as finite magnitudes are concerned) for Euclid's celebrated 12th Axiom, preceded by, and dependent on, the Axiom tacitly assumed by him in his Book X, Prop. 1, and also assumed, I believe, by every subsequent writer who has attempted to prove his 12th Axiom. My proof is borrowed, with some slight alterations, from Cuthbertson's 'Euclidean Geometry.'

One advantage, in thus separating the Propositions into two classes, is that it calls attention to the very remarkable[Pg x] and interesting fact that the Theorem "There is a Triangle whose angles are together not-greater than two right angles" is actually provable without any disputable Axiom whatever. If only it could be proved, with equal ease, that "there is a Triangle whose angles are together not-less than two right angles"! But alas, that is an ignis fatuus that has never yet been caught! The man, who first proves that Theorem, without using Euclid's 12th Axiom or any substitute for it, will certainly deserve a place among the world's great discoverers.

I take this opportunity of replying to one or two criticisms, which have been published, on the Second Edition—earnestly assuring the writers of those criticisms that, in treating the questions at issue between us from a not-wholly-solemn point of view, I have been actuated by no feeling of disrespect towards them, but simply from the wish to lighten a subject, naturally somewhat too heavy and sombre, and thus to make it a little more palatable to the general Reader.

At p. 12 of the 2nd Edition, the enunciation of Prop. VI (which re-appears, in a modified form, at p. 34 of the 3rd Edition) stood thus:—

"If the vertical angle of a Sector of a Circle be divided by radii into equal angles, thus forming equal Sectors; and if the chord of each such Sector be not less than the radius of the Circle: the original Sector is not less than times the Triangle cut off from it by its chord." My controversy with Nature, on this enunciation, will be best given in the form of a dialogue. (Of course I quote verbatim.)

Nature. (Dec. 6, 1888.) "How are the figures to be constructed, if n be greater than 2?"

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Author. (In the Preface to the 2nd Edition, at p. x.) "Well, suppose n were equal to 4: i.e. we have to divide the vertical angle into 24 equal parts. Bisect it: that gives halves. Bisect the halves: that gives quarters. Bisect again: that gives eighths. Bisect

once more: that gives sixteenths. Voila tout!"

Nature. (June 13, 1889.) "Shade of Euclid! Who knows not such things? We admitted the same, but stated that our difficulty in the construction was the condition imposed in the enunciation: viz., 'the chord of each such sector not less than the radius of the circle.' Take Mr. Dodgson's illustration of a sixteenth: this would necessitate that the original angle should be at least 960° ... we ... have further noted that no one of the chords in Mr. Dodgson's figures is even equal to the radius."

Author. "What you call 'the condition imposed' is introduced with an 'if': it is merely an hypothesis: all I undertake to prove is that, if certain things were true, certain other things would be true. Surely I need not remind you that the validity of a Syllogism is quite independent of the truth of its Premisses! 'I have sent for you, my dear Ducks', said the worthy Mrs. Bond, 'to enquire with what sauce you would like to be eaten?' But we don't want to be killed!' cried the Ducks. 'You are wandering from the point' was Mrs. Bond's perfectly logical reply. So here. 'I beg you to observe, my dear Nature, that, if the chord of each Sector were not less than the radius, the logical result would be so-and-so.' 'But the chord is less than the radius!' you cry. All I need say, in reply, is 'You are wandering from the point.'"

"But I will be generous, and will say more. I take exception to two assertions of yours. Remember our[Pg xii] logical stand-point. We may use Euclid's Axioms, all but the last; and his Propositions as far as I. 28. Now be good enough to prove to me, with this machinery, first, that my hypothesis 'necessitates that the original angle should be at least 960°'; secondly, that 'no one of the chords' in my Figure 'is even equal to the radius.' Your logical position is, I fear, this. You dispute the validity of a certain argument, on the ground that its premisses are false. My reply is, first, that you cannot prove them false; and secondly, that, even if you could, it wouldn't affect the question!"

At p. 19 of the 2nd Edition, the new Axiom, on which my Theory rests, (which re-appears, in a modified form, at p. 14 of the 3rd Edition), stood thus:—"In every Circle, the inscribed equilateral Hexagon is greater than any one of the Segments which lie outside it." My controversy with the Athenæum, on this Axiom, shall also be given in the form of a dialogue.

Athenæum. (Oct. 27, 1888.) "... a stronger objection, in our opinion, is the implied assumption of the possibility of the inscribed equilateral Hexagon, a possibility which is not demonstrated till we reach the fifteenth Proposition of Euclid's fourth book."

Author. (In the Preface to the 2nd Edition, at p. xi). "But does it need demonstrating? May we not assume (1) that the Magnitude 'four right angles' contains 6-6ths of itself; (2) that it is theoretically possible to draw radii dividing it into these 6-6ths? Once grant me this, and I ask no more. I have then the logical right to join the ends of these radii, and to prove (by Euc. I. 4) that the chords are equal."

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Athenæum. (Oct. 5, 1889.) "We objected that it was not consistent with the spirit or practice of Euclid's reasoning to assume the 'theoretical possibility' of a regular

Hexagon inscribed in a Circle, without first proving that such a figure could be actually constructed from his three postulates. Euclid's restrictions may be arbitrary, unnecessary, cramping, vexatious, absurd—indeed, we think they deserve these and many other epithets—but there they are, and, if Mr. Dodgson accepts them, he is bound to keep his assumptions within the boundaries which they prescribe."

Author. "You're particular to a shade (as Scrooge said to Marley's ghost): however, I'll do what I can to oblige you. I presume you will be satisfied if I can, without using more of Euclid than his first 28 Propositions, construct an angle which shall be 1-6th of 4 right angles? Very good. First, then, with the help of his arbitrary Prop. I, I construct an equilateral Triangle. Next, by his unnecessary Prop. IX, I draw the bisectors of 2 of its angles. Next, by his cramping Post. 1, I join their point of intersection to the third vertex. Next, by his vexatious Prop. IV, I prove the 3 angles, whose common vertex is this point, to be equal. From which I draw the absurd conclusion that each of them is 1-3rd (and that therefore its half is 1-6th) of 4 right angles. How does that strike you?"

Another objection, to this same Axiom, appeared in the Academy for Feb. 9, 1889, viz. "What the Axiom practically assumes is the existence of similar figures." Permit me to reply, to this Reviewer, as follows:—"In what sense do you use the word 'similar'? In Euclid's, no doubt. That is to say, you charge me with assuming that, if the Circle[Pg xiv] and its inscribed Hexagon were supposed to expand, the magnitude of each angle, and the ratio subsisting between the sides which contain it, would remain constant? The 'ratio' part of the question we may set aside at once: there is no doubt that, since the figure continues to be equilateral, the ratio continues to be a ratio of unity: hence, if I needed this assumption (which I don't), I should have a perfect right to make it. All, that remains for discussion, is the assumption, which you say I have made, that each angle of the expanding Hexagon remains constant in magnitude. Will you, then, be kind enough to point out, first, where the need for any such assumption arises; secondly, where I have made any such assumption? For myself I cannot in the least see why, in estimating the area of the Hexagon, I should trouble myself about the size of its angles."

Let me take this opportunity of pointing out, once more, that not one Proposition in this Treatise depends, in the slightest degree, on the speculations about Infinities, &c., which occur in the Appendices.

The one merit, the one novelty, of my Theory (if it has any merit, or any novelty) is that, while every other Theory (that I have seen), which attempts to supersede Euclid's 12th Axiom, introduces the ideas of Infinities and Infinitesimals, mine dispenses wholly with their aid, and deals with nothing but what is, by universal consent, absolutely within the field of Human Reason.

C. L. D.

Ch. Ch., Oxford.

August, 1890.

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INTRODUCTION.

It may well be doubted whether, in all the range of Science, there is any field so fascinating to the explorer—so rich in hidden treasures—so fruitful in delightful surprises—as that of Pure Mathematics. The charm lies chiefly, I think, in the absolute certainty of its results: for that is what, beyond almost all mental treasures, the human intellect craves for. Let us only be sure of something! More light, more light! Ἐν δὲ φάει, καὶ ὀλέσσον. 'And, if our fate be death, give light and let us die!' This is the cry that, through all the ages, is going up from perplexed Humanity, and Science has little else to offer, that will really meet the demands of its votaries, than the conclusions of Pure Mathematics. Most other Sciences are in a state of constant flux—the precious truths of one generation being smiled at as paradoxes by the second generation, and contemptuously swept away as childish nonsense by the third. If you would see a specimen of the rapidity of this process of decomposition, take Biology for a sample: quote, to any distinguished Biologist you happen to meet, some book published thirty years ago, and observe his pitying smile!

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But neither thirty years, nor thirty centuries, affect the clearness, or the charm, of Geometrical truths. Such a theorem as 'the square of the hypotenuse of a right-angled triangle is equal to the sum of the squares of the sides' is as dazzlingly beautiful now as it was in the day when Pythagoras first discovered it, and celebrated its advent, it is said, by sacrificing a hecatomb of oxen—a method of doing honour to Science that has always seemed to me slightly exaggerated and uncalled-for. One can imagine oneself, even in these degenerate days, marking the epoch of some brilliant scientific discovery by inviting a convivial friend or two, to join one in a beefsteak and a bottle of wine. But a hecatomb of oxen! It would produce a quite inconvenient supply of beef.

Now this field of Mathematical research, with all its wealth of hidden treasure, is all too apt to yield nothing to our research: for it is haunted by certain ignes fatui—delusive phantoms, that float before us, and seem so fair, and are all but in our grasp, so nearly that it never seems to need more than one step further, and the prize shall be ours! Alas for him who has been turned aside from real research by one of these spectres—who has found a music in its mocking laughter—and who wastes his life and energy in the desperate chase!

One of these—the chief one, I think—is the old old problem of 'Squaring the Circle,' which has certainly wasted many a human life. Whether it has actually driven any one mad, I know not—most of its victims were, I fancy, partly crazed before they entered on the quest—but it clearly has the power of demolishing such slender reasoning powers as they may ever have chanced to possess.

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With two of these misguided visionaries I have myself corresponded.

The first who addressed me filled me with a great ambition—to do a feat I had never yet heard of as accomplished by man, namely, to convince a 'Circle-Squarer' of

his error! The value my friend had selected for '' was not an original one—being 3·2: but the enormous error, beginning as early as the first decimal place, tempted one with the idea that it could be easily demonstrated to be an error. I should think more than a score of letters were interchanged before I became sadly convinced that I had no chance. What man could still hope on, after receiving such a rebuff as the following? "You persuade yourself," so my friend wrote, "that you have made your circumscribed polygon equal to the circle, which you know cannot be, and have thereby pushed the quadrant beyond 90°, valuing the circumference at 360°." I meekly begged to be referred to the actual words in which I had advanced this startling assertion: but I never succeeded in getting the quotation verified.

My second 'Circle-Squarer' went to work in quite another fashion. His object was not so much to obtain an arithmetical value for ',' as to construct a geometrical straight Line which, given the radius, should exhibit to the eye the actual length of the circumference. His diagram was a most imposing one—Triangles and Parallels were interlaced in bewildering profusion—and it used up no less than 23 letters of the alphabet. Some of the Lines had arithmetical values assigned to them: and there was one value, '1·8879020478639098461 &c.', which for a long time baffled all my endeavours to guess how in the world he had invented it. Of course one might have taken exception[Pg xviii] to such a construction at the very outset, and have said "I will admit the possibility of constructing a Line, which shall bear to the unit-line any arithmetical ratio you like, so long as you express it as an exact decimal: but what can I do with your '&c.'?" But his was not the kind of mind to which the geometrical construction of an '&c.' presents any difficulty. At length, after many failures, I chanced on the discovery that this portentous number was of the decimal part of '.' After this it was no wonder, considering that, in the course of construction, he had taken of this Line, and afterwards divided by 10, that the resulting Line, added to 3 times the unit-Line, was triumphantly proved to represent ''! I ventured to ask if this was the way he had obtained the long decimal quoted above, namely, by multiplying the decimal part of '' by , and received the courteous reply "your suggestion is perfectly correct"!

Another ignis fatuus—though not numbering so many victims as the 'Quadrature of the Circle'—is 'the Trisection of an Angle' (that is, its trisection by Euclid's machinery).

And yet another ignis fatuus—the one with which the following treatise is concerned—is the attempt to dispense with Euclid's celebrated 12th Axiom.

I may as well state briefly what the feat actually is, which Mathematicians have been vainly trying, since Euclid's day, to perform.

In I. 27, 28, he proves (so far, without invoking the aid of any doubtful Axiom) that "two Lines, which are equally inclined to a certain transversal (whether by making a pair of alternate angles equal, or an exterior equal to its interior opposite angle, or two interior, on the same side of[Pg xix] the transversal, together equal to two right angles), will never meet."

Next, in logical order, comes his 12th Axiom, viz. that "two Lines, which are unequally inclined to a certain transversal (he only names the case where they make two interior angles together less than two right angles, but he might fairly have included the others), will meet." This Axiom, as I hope to prove in Appendix III, is only partially, and not universally, true.

Next, in I. 29, he proves (with the aid of this Axiom, of which it is what De Morgan calls the 'contranominal') the partially-true Theorem that "two Lines, which never meet, are equally inclined to any transversal."

And from this, in I. 32, he proves that "the three angles of a Triangle are together equal to two right angles."

These are only specimens of a set of Theorems which can be proved when once Axiom 12 is granted (e.g. there are several about 'equidistantial Lines,' which Euclid has altogether ignored): but they are all so connected as to follow easily from these.

Now the great difficulty, which besets this subject, is that Euclid's Axiom (this, I think, is universally admitted) is not axiomatic—the intellect has not yet occurred, among that species of Vertebrates which may be defined as 'bimanous bipeds,' which accepts it as a genuine Axiom—and the great question to be answered is "can a better Axiom be found?"

In Appendix IV, I will mention some of the substitutes that have been suggested, and will give some account of the 'outlook' in the direction of the new Axiom I have chanced on. In this place it will suffice, first, to explain what the task is that the long-desiderated Axiom has to[Pg xx] perform, and secondly, to state the grounds on which I claim acceptance for my Axiom.

First, then, what is 'the coming Axiom' expected to do for us?

It will be convenient to divide the whole class, of Theorems needing proof, into two sub-classes—one including those which are universally true: the other those which are only partially true—the error, if any, being infinitesimal when compared with the Magnitudes with which the Theorem is concerned.

Euc. I. 32 is a specimen of the one kind, and Euc. I. 29 of the other.

In proving the latter class, no substitute for Euclid's Axiom has yet been suggested, that I know of, which does not suffer from the same defect as Euclid's Axiom—the being only partially, and not universally, true—and which does not, if we attempt to modify the language so as to remedy this defect, in some way lead us into the bewildering region of Infinities and Infinitesimals.

But the former class can, as I believe, be more easily proved. This is what I attempt in the following treatise—which owes its inspiration to a sudden thought (it occurred to me some two months ago) that it might be possible to prove Euc. I. 32 without getting mixed up with those spectral Infinities.

Moreover, it is quite possible to bring into this class all that is valuable in Euc. I. 29. Regarding the 'separateness' of the Lines merely as a link between Props. 27, 28, and 29, we may combine the three into one grand Theorem, thus:—"Two Lines, which

are equally inclined to a certain transversal, are so to every transversal." This Theorem, as well as Euc. I. 32, I prove in the following[Pg xxi] treatise. But the feat of proving them, without assuming any new Axiom at all, is at present beyond my grasp. Like the goblin 'Puck,' it has led me "up and down, up and down," through many a wakeful night: but always, just as I thought I had it, some unforeseen fallacy was sure to trip me up, and the tricksy sprite would "leap out, laughing ho, ho, ho!"

And now, to come to the real gist of this over-long Preface—however, nobody ever reads a Preface, so really it does not matter—am I not right in thinking that, on mere inspection of this diagram, any sane intellect will be ready to grant that "in any Circle, the inscribed Tetragon is greater than any one of the Segments that lie outside it"?

I shall be told, no doubt, that this is too bizarre and unprecedented an Axiom—that it is an appeal to the eye, and not to the reason. That it is somewhat bizarre I am willing to admit—and am by no means sure that this is not rather a merit than a defect. But, as to its being an appeal to the eye, what is "two straight Lines cannot enclose a space" but an appeal to the eye? What is "all right angles are equal" but an appeal to the eye?

In all Axioms, where an appeal is made to the eye on a question of magnitude, we shall find, I think, that the whole region of certainty and probability may be roughly mapped out into three districts—an out-lying district of certainty in one direction, a similar one of certainty in the opposite direction, and a middle district of probability—the boundaries being shadowy and liable to be shifted hither[Pg xxii] and thither according to the fancies or prejudices of each individual mind.

Permit me to illustrate this by an example taken from ordinary life.

You enter a room, where there is a book-case containing (say) five shelves, and your eye wanders carelessly along a shelf, making a rough estimate of the number of books in it. Now shut your eyes, and try to guess how many books there are altogether. Your hasty reckoning of one shelf gave a total (say) of 19 or 20, you are not sure which: so you feel safe in saying "I think there are about a hundred." "Are you certain," I ask, "that there are more than fifty?" "Quite certain," you reply. "And also certain that there are less than a hundred and fifty?" "Quite certain," you repeat. Here, then, are the three districts. The numbers up to 50 are certainly too small; the numbers over 150 are certainly too great; the intermediate numbers contain some doubtful ones—the most doubtful being very near 100—and this doubt shades off into certainty as we approach either of the out-lying districts. You would not risk five shillings on the chance of the true number being under 100, or on the chance of its being over 100, but you would feel quite at your ease, if told that you would forfeit a thousand pounds, in case the number turned out to be under 50, or over 150.

Another objection, that has already been raised to my Axiom, and so will probably be raised again, and which I may as well meet here by anticipation, is that, on the supposition of Euclid I. 32 not being true, it may be proved that this relationship of magnitude, between the Tetragon and the Segment, changes as the Circle increases, until, with an infinitely great Circle, the Tetragon may[Pg xxiii] actually be proved to

be less than the Segment! This phenomenon, however, does not appal me so much as might be expected: for I have often observed it to occur that, when Theorem logically leads to Theorem , then, on the supposition of Theorem not being true, it may be proved that Theorem also is not true. (The second sequence is, in fact, what De Morgan calls the 'contranominal' of the first.) Hence this objection, if worth anything, proves too much: to dispute the validity of an argument, on the ground that, if it were valid, its contranominal would also be valid, is to upset the whole edifice of Logic itself: and, if you tell me, on such grounds as these, that I cannot prove what I assert, I may fairly retort upon you, that you cannot prove anything at all! You have destroyed the only machinery available for the purpose, and must henceforth dispense with all Logical methods, and console yourself with the cynical American adage "There's nothing true: and there's nothing new: and it don't signify!"

To return to our Tetragon. It really contains the area of the Segment a little over 7 times. Hence anybody, I should suppose, would be ready to say "I am certain it contains the Segment more than twice: and I am equally certain it does not contain it twelve times." In guessing the actual number, observers would greatly differ: some might guess 4, others 10: but all would agree in putting it above 2. And now see how modest is the demand of my Axiom! Merely that you will find room in the Tetragon for one single Segment! If that is not a matter of certainty, is anything certain in this world of ours?

I have yet one more arrow in my quiver: let me shoot it, and have done. If the gentle reader feels any the[Pg xxiv] smallest demur to granting me that once this Tetragon is greater than the Segment lying below it, will he grant me that twice it will suffice? Or four times it? Or eight times it? He may go on doubling as long as he likes, and, so long as he keeps among finite numbers, he will have granted me all I need for a logical proof (which will be found in Appendix I) of Euc. I. 32. Surely he will not need to go into the Infinities? And may I add, in conclusion, that, if any gentle Reader be found, who thinks it just possible to squeeze 512 of these Tetragons into the Segment, but is willing to allow that no amount of skilful packing will dispose of 1024 of them—it will give me real satisfaction to be supplied with that gentle Reader's name and address?

C. L. D.

Ch. Ch., Oxford.

July, 1888.

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A NEW THEORY OF PARALLELS.

Book I.

Certain universally-true Propositions, provable from genuine Axioms.

Definitions.

1.

The sum of the angles of a Triangle is called its 'amount.'

2.

Any angular magnitude is called a 'possible amount,' if there be a Triangle whose 'amount' is equal to it: but, if there be no such Triangle, it is called an 'impossible amount.'

3.

If any such angular magnitude vary continuously: whenever it changes from a 'possible amount' to an 'impossible amount,' it is said to pass from a 'possible region,' to an 'impossible region': and vice versâ.

[Pg 2]

4.

If there be two fixed angular magnitudes such that the varying magnitude, while it continues between them, is always a 'possible amount,' but becomes an 'impossible amount' when it passes beyond them: they are called the 'superior limit,' and the 'inferior limit,' of the 'possible region' which lies between them.

Axioms.

[Pg 3]

1.

If a Magnitude change from one value to another; and if, in doing so, it vary continuously; and if a certain value, intermediate to its first and last values, be selected: the Magnitude must, at some moment during the process of change, have that selected value.

2.

If two or more Magnitudes be such that, whenever any one of them varies, it varies continuously: then, whenever their sum varies, it varies continuously.

3.

If , , be two Lines intersecting at ; and if be turned about , remaining stationary: each of the angles at varies continuously.

4.

If , , be two intersecting Lines; and if be turned about , remaining stationary: then, so long as the Lines continue to intersect, each of the 4 angles at the Point of intersection varies continuously.

Propositions.

[Pg 4]

PROP. I. Theorem.

If a Pair of Lines make, with a certain transversal, either (1) a pair of alternate angles equal, or (2) an exterior angle equal to its interior opposite angle on the same side of the transversal, or (3) a pair of interior angles on the same side of the transversal supplementary: they will make, with that transversal, (4), each pair of alternate angles equal, and (5) each of the four exterior angles equal to its interior opposite angle on the same side of the transversal, and (6) each pair of interior angles on the same side of the transversal supplementary.

This Proposition is easily deduced from Euc. I. 13, 15.

Definitions (continued).

5.

Such a Pair of Lines may be said to be 'equally inclined' to that transversal.

PROP. II. Theorem.

[Pg 5]

If two isosceles Triangles have equal bases but unequal sides: that Triangle, which has the greater sides, has the greater area.

Let the Triangles be set on the same base, and call them , . And let , , be respectively greater than , .

Now cannot fall within the Triangle , or upon either of its sides; for then , would be less than , ;

[Euc. I. 21.

neither can it fall on ; for then , would be equal to , ;

neither can intersect , nor intersect ; for, in either case, if were joined, there would be two Triangles, on the same base , having their coterminous sides equal; which is impossible;

[Euc. I. 7.

, fall outside the Triangle ;

Triangle is greater than Triangle .

Therefore, if two isosceles Triangles &c.

Q.E.D.

[Pg 6]

PROP. III. Problem.

Given a certain angle; and given that any isosceles Triangle, whose vertical angle is not-greater than the given angle, has its base not-greater than either of its sides: to describe, on a given base, an isosceles Triangle having each base-angle equal to the given angle.

Let be given base.

At , in Line , make angle equal to given angle, making ; and join .

Then, by hypothesis, is not-greater than : i. e. it is either equal to it, or less than it.

First let be equal to . (Fig. 1.)

Then Triangle is equilateral: i. e. it is an isosceles Triangle, on given base , and having each base-angle equal to given angle.

Q.E.F.

Secondly, let be less than . (Fig. 2.)

Then angle is less than angle .

[Euc. I. 18.

At make angle equal to angle .

[Euc. I. 23.

Then Triangle is isosceles, and is on given base , and has each base-angle equal to given angle.

Q.E.F.

[Pg 7]

Corollary.

The isosceles Triangle, so described, has its vertical angle not-less than either of its base-angles.

For, in Fig. 1, ;

.

Q.E.D.

Again, in Fig. 2, ;

.

But angle is greater than angle ;

[Euc. I. 16.

and angle is less than angle ;

.

Q.E.D.

[Pg 8]

PROP. IV. Theorem.

Either all Triangles have the same 'amount'; or else, if , be two 'possible amounts,' that is, 'amounts' belonging to existing Triangles, then any 'amount,' intermediate to and , is also 'possible.'

If all Triangles have the same amount, the Proposition is true. If not, let , be two Triangles whose amounts are different. Call their 'amounts' ', .' And let the two Triangles be placed so as to have a common vertex at , and their bases in the same straight Line.

Now Triangle may be converted into Triangle by making the point, where intersects , move from to , remaining stationary.

And, during this process, the angle at will vary continuously,

[Ax. 4.

and the angle, at the point where the revolving Line intersects , will vary continuously;

[Ax. 5.

the sum of these angles will, if it vary at all, vary continuously.

[Ax. 3.

Similarly, Triangle may be converted, first, into Triangle , by making the point, where intersects , move from to , remaining stationary, and then into Triangle , by making the point, where intersects , move from to , remaining stationary.

And, during the whole process, the 'amount' of the changing Triangle will, if it vary at all, vary continuously.

Hence, in changing from the value to the value , it must pass through all intermediate 'amounts': i. e. all intermediate 'amounts' are 'possible.'

[Ax. 2.

Therefore either all Triangles have &c.

Q.E.D.

Corollaries.

[Pg 9]

1.

Among angular magnitudes there is one, and only one, 'possible region.'

2.

This 'possible region' either consists of one single angular magnitude, such that it, and it alone, is a 'possible amount'; or it consists of a continuous series of angular magnitudes, lying between 2 'limits,' which 2 limits are such that any magnitude, lying between them, is a 'possible amount,' and any magnitude, lying outside them, is an 'impossible amount.'

PROP. V. Theorem.

[Pg 10]

The angles of any Triangle are together less than three right angles.

Let a right angle be represented by '.'

Now any 2 of the angles of a Triangle are together less than ;

[Euc. I. 17.

, adding together the 3 pairs which may be taken, the 3 angles of a Triangle, taken twice over, are together less than ;

, taken once only, they are together less than .

Therefore the angles of any Triangle &c.

Q.E.D.

PROP. VI. Theorem.

There is a Triangle whose angles are together not-greater than two right angles.

If we deny this, we must assert that any 'amount' is greater than .

Let this be our First Hypothesis.

Now any 'amount' is less than .

[Prop. 5.

Hence the 'possible region' lies below ; i. e. it has a 'superior limit.'

Now let a 'possible amount' be selected, more than half-way from to the 'superior limit' of this region; and call it '.' Then it is evident that will lie above this limit, and will therefore be an 'impossible amount.'

[Pg 11]

Now any Triangle, whose 'amount' is , must be either obtuse-angled, or right-angled, or acute-angled.

Hence we must assert that there is either an obtuse-angled Triangle, or a right-angled Triangle, or an acute-angled Triangle, whose 'amount' is .

Let these be our Second, our Third, and our Fourth Hypotheses.

Call the Triangle '.'

First, let it be obtuse-angled; and let be the obtuse angle.

At Point , in Line , make angle equal to angle , making equal to ; and join and .

Then Triangle is equal, in all respects, to Triangle ;

[Euc. I. 4.

its 'amount' = ;

also 'amount' of Triangle is, by our First Hypothesis, greater than ;

'amounts' of the 3 Triangles are together greater than .

But these make up 'amount' of Triangle , plus angles about , which = ;

[Euc. I. 13. Cor.

'amount' of Triangle , plus , is greater than ;

this 'amount,' alone, is greater than ; which is absurd, since the latter lies above the 'superior limit,' and is therefore an 'impossible amount.'

Hence our Second Hypothesis is false; i.e. no obtuse-angled Triangle can have the amount .

[Pg 12]

Secondly, let it be right-angled; and let be the right angle.

At Point , in Line , make angle equal to angle , i. e. equal to ; and make ; and join .

Then , , are in one straight Line.

[Euc. I. 14.

Also Triangle is equal, in all respects, to Triangle ;

[Euc. I. 4.

its 'amount' = ;

'amounts' of the 2 Triangles together = .

But these make up 'amount' of Triangle , plus angles at , which = ;

'amount' of Triangle , plus ;

this 'amount,' alone, = ; which is absurd.

Hence our Third Hypothesis is false; i.e. no right-angled Triangle can have the amount .

Thirdly, let it be acute-angled.

Bisect at ; join , and produce it to , making ; and join .

Now angles , , are equal, being vertical;

[Euc. I. 15.

Triangles , , are equal in all respects;

[Euc. I. 4.

angle = angle ; and angle = angle ;

'amount' of Triangle = that of Triangle .

Again, angle = angle ;

angles , together = angle ;

they are together less than ;

angle is, by our First Hypothesis, greater than ;

i. e. Triangle is obtuse-angled;

its 'amount' cannot be ;

'amount' of Triangle cannot be .

Hence our Fourth Hypothesis is false; i.e. no acute-angled Triangle can have the 'amount' .

Hence, no Triangle can have this amount.

Hence our First Hypothesis, that any 'amount' is greater than , is false.

Therefore there is a Triangle &c.

Q.E.D.

Corollary.

[Pg 13]

The 'possible region' does not lie wholly above two right angles.

Book II.

Certain universally-true Propositions, not provable from genuine Axioms, but provable if the following Axiom be accepted.

[N.B. This Axiom cannot claim to be more than a 'Quasi-Axiom,' i. e. one whose self-evident character is disputable.]

Axioms.

[Pg 14]

1.

In any Circle, the inscribed equilateral Tetragon is greater than any one of the Segments which lie outside it.

Note.—If, in any Circle, 2 Diameters be drawn at right angles to each other, and their extremities joined, the joining Lines will, by Euc. I. 4 be equal to each other. Hence the Figure, thus formed, will be an inscribed equilateral Tetragon. (It will also be equiangular; but that is of no importance for our present purpose.)

PROP. I. Theorem.

[Pg 15]

An isosceles Triangle, whose vertical angle is one-eighth of a right angle, has its base less than either of its sides.

Let be an isosceles Triangle, whose vertical angle at is one-eighth of a right angle.

It shall be proved that is less than .

If we deny this, we must assert that is not-less than .

Let this be our Hypothesis.

Construct 7 more Triangles , &c., equal to . With centre , and distance , describe quadrant passing through , , &c. (See Note.) And join , , , , , , .

Note.—The Reader is requested to imagine chords drawn to the arcs , , &c.

[Pg 16]

Then Triangle is one-fourth of an equilateral Tetragon inscribed in the Circle.

Hence, 4 times this Triangle is greater than Segment .

[II. Ax. 1.

Because angle is greater than angle , and that angle is less than angle ;

angle is greater than angle ;

is greater than .

[Euc. I. 19.

Similarly, is greater than .

Hence, on our Hypothesis, and are both of them greater than .

Also, is greater than ;

Triangle is greater than Triangle ;

[I. Prop. 2.

to each of these add Triangle ;

Figure is greater than twice Triangle .

Again, is greater than ;

Triangle is greater than Triangle ;

[I. Prop. 2.

Triangles , are together greater than Figure ;

to each of these add Figure ;

Figure is greater than twice Figure , i.e. greater than 4 times Triangle .

Again, is not-less than ;

Triangle is not-less than Triangle ;

[I. Prop. 2.

Triangles , , , are together not-less than Figure ; i.e. they are together greater than 4 times Triangle ;

, a fortiori, Segment is greater than 4 times Triangle .

But this is absurd, since it has been already proved less than 4 times this Triangle.

Hence our Hypothesis, that is not-less than , is false; i.e. is less than .

Therefore an isosceles Triangle &c.

Q.E.D.

Corollary.

[Pg 17]

Hence, by Book I, Prop. III, it is possible to describe, on a given base, an isosceles Triangle having each base-angle equal to one-eighth of a right angle.

PROP. II. Theorem.

[Pg 18]

The angles of any Triangle are together not-less than one-eighth of a right angle.

Let one-eighth of a right angle be represented by '.'

Let be a Triangle; it shall be proved that its 'amount' is not-less than .

If we deny this, we must assert that its 'amount' is less than .

Let this be our Hypothesis.

Hence each of its angles is less than .

On describe an isosceles Triangle having each base-angle equal to ;

[II. Prop. 1. Cor.

hence , , must lie within this Triangle;

i.e. Triangle must lie within it;

angle is less than angle ;

[Euc. I. 21.

i.e. less than ;

but angle is not-less than angle ;

[I. Prop. 3. Cor.

i.e. not-less than ; which is absurd.

Hence our Hypothesis, that 'amount' of Triangle is less than , is false; i.e. it is not less than .

Therefore the angles of any Triangle &c.

Q.E.D.

Corollary.

The 'possible region' does not extend below one-eighth of a right angle.

PROP. III. Theorem.

[Pg 19]

There is a Triangle whose angles are together not-less than two right angles.

If we deny this, we must assert that any 'amount' is less than .

Let this be our First Hypothesis.

It shall be proved that, if we assert this, we must also assert that any 'amount' is not-less-than , where '' represents one-eighth of a right angle.

If we deny this, we must assert that there is an 'amount' less than .

Let this be our Second Hypothesis.

Now we know that the 'possible region' does not extend below ;

[II. Prop. 2. Cor.

i.e. it has an 'inferior limit.'

Let a 'possible amount' be selected, more than half-way from to this 'inferior limit,' and call it '.' Then it is evident that will lie below the 'inferior limit,' and will therefore be an 'impossible amount.'

Let a Triangle be taken, whose 'amount' is ;

any one of its angles, which is not-greater than either of the others, is not-greater than ; i.e. is less than .

Call this angle '.'

Now one, at least, of the remaining angles must be acute.

[Euc. I. 17.

Call this ''; and call the third angle .

[Pg 20]

Let 2 such Triangles, and , be taken; and let them be so placed that their -vertices coincide and their -sides lie in one straight line; and join .

On describe an isosceles Triangle , having each base-angle equal to .

[II. Prop. 1. Cor.

Now each of the angles , , is less than .

Lines , , will fall within angles , ;

i.e. Figure will fall within Triangle .

Join , DC′.

Now 'amounts' of Triangles , , together = , and those of the other 4 Triangles are, by our First Hypothesis, together less than ;

'amounts' of all 6 Triangles are together less than ;

but these make up 'amount' of Triangle , plus angles at , , , which together = ;

'amount' of Triangle , plus , is less than .

Now we know that is not-greater than ;

, adding these inequalities, 'amount' of Triangle , plus , is less than ;

this 'amount,' alone, is less than ; which is absurd, since the latter lies below the 'inferior limit,' and is therefore an 'impossible amount';

one of our two Hypotheses must be false;

i.e. either there is an 'amount' not-less than , or else any 'amount' is not-less than .

Suppose we maintain our First Hypothesis: then we must abandon our Second; i.e. we must admit that any 'amount' is not-less than .

It shall be proved that, in this case, we must also admit that any 'amount' is not-less than .

For, if we deny this, we must assert that there is an 'amount' less than .

Let this be our Third Hypothesis.

Then, in the above proof, and may be replaced by and , and a similar absurdity will follow.

Hence either our First or our Third Hypothesis must be false;

i.e. either there is an 'amount' not-less than , or else any amount is not-less than .

A similar proof will hold for ; and then for .

Hence, either there is an 'amount' not-less than , or else any amount is not-less than .

But .

Hence the second clause of this alternative contains the first.

Hence the first clause must be true.

That is, there is a Triangle &c.

Q.E.D.

Corollary.

[Pg 21]

The 'possible region' does not lie wholly below two right angles.

PROP. IV. Theorem.

[Pg 22]

There is a Triangle whose angles are together equal to two right angles.

For the 'possible region' does not lie wholly above ;

[I. Prop. 6. Cor.

neither does it lie wholly below ;

[II. Prop. 3. Cor.

it includes .

[I. 4. Cor. 2.

That is, there is a Triangle &c.

Q.E.D.

PROP. V. Theorem.

There is a quadrilateral Figure which is 'rectangular,' that is, which has all its angles right angles.

Let be a Triangle whose 'amount' = .

[II. Prop. 4.

At make angle equal to angle , and angle equal to angle ;

hence angles , , , together = ;

, , are in a straight Line.

[Euc. I. 14.

Bisect at ; from draw perpendicular to ; from cut off equal to ; and join .

, in Triangles , , , , are respectively equal to , , and angle to angle , [Euc. I. 4.

the Triangles are equal in all respects;

angle is a right angle;

and angle ;

angles , , together = angles , ; i. e. together = ;

, are in a straight line;

[Euc. I. 14.

is a common perpendicular to Lines , .

Similarly, by bisecting at , it may be proved that is a common perpendicular.

Hence Figure is rectangular.

Therefore there is a quadrilateral Figure &c.

Q.E.D.

Definition.

[Pg 23]

A rectangular quadrilateral Figure may be called a 'Rectangle.'

PROP. VI. Theorem.

[Pg 24]

The opposite sides of a Rectangle are equal.

Let be a Rectangle; and let it be reversed so that , , may change places.

Then will lie along , and along .

Now, if were not equal to , , , would not change places, but would take new positions, as ;

hence exterior angle would be greater than interior opposite angle ;

[Euc. I. 18.

but they are also equal, being right angles; which is absurd;

.

Similarly it may be proved that .

Therefore the opposite sides &c.

Q.E.D.

PROP. VII. Theorem.

There is a Pair of Lines, each of which is 'equidistant' from the other, that is, is such that all Points on it are equally distant from the other Line.

Let be a Rectangle; and let a vertical Line be supposed, first to coincide with , and then to move along , continuing always at right angles to it, till it reaches some intermediate position .

Now, if its top be not now on , it must have either dropped below it or risen above it.

First, let it be supposed to have dropped below it: and join , .

Then, if the Figure be reversed, and applied to the same base, it is evident that and will exchange places;

angle , i.e. it is less than ;

Similarly angle is less than ;

but angle is less than .

angles at are together less than ; which is absurd;

[Euc. I. 13. Cor.

top of vertical Line has not dropped below .

Similarly it may be proved that it has not risen above .

Hence it moves along ; i. e. it describes a straight Line, and will evidently continue to do so, however far the vertical Line move, either way, along .

Therefore there is a Pair of Lines &c.

Q.E.D.

Corollaries.

[Pg 25]

1.

If a Pair of Lines have a common perpendicular: each of them is equidistant from the other.

Corollary 2.

[Pg 26]

It is possible to form a Rectangle of any given width and height.

For, in the above Pair of horizontal equidistant Lines, 2 common perpendiculars may be drawn, at a given width apart; and the Figure, so formed, will be a Rectangle; and its sides will be a Pair of vertical equidistant Lines, which may be treated in the same way.

PROP. VIII. Theorem.

The angles of any Triangle are together equal to two right angles.

Let be a Triangle, so placed that each base-angle is acute; from draw perpendicular to ; and make Rectangles , .

[II. Prop. 7. Cor. 2.

has its opposite sides equal,

[II. Prop. 6.

, in Triangles , , the sides of the one are respectively equal to the sides of the other;

;

[Euc. I. 8.

Similarly angle ;

the angles of the Triangle together = the angles , , ; i. e. together = .

Therefore the angles of any Triangle &c.

Q.E.D.

PROP. IX. Theorem.

[Pg 27]

A Pair of Lines, which are equally inclined to a certain transversal, are so to any transversal.

Let , , be equally inclined to transversal ; and let be any other transversal.

Join .

Now 'amounts' of Triangles , , together = .

[II. Prop. 8.

But these make up angles of Figure ;

angles of Figure together = ;

but angles , together = ;

[I. Prop. 1.

angles , , together = ;

, , are equally inclined to .

[I. Prop. 1.

Therefore a Pair of Lines &c.

Q.E.D.

Axioms (continued).

2.

If two homogeneous magnitudes be both of them finite: the lesser may be so multiplied, by a finite number, as to exceed the greater.

PROP. X. Theorem.

[Pg 28]

If a Pair of Lines make, with a certain transversal, two interior angles, on the same side of it, which are together less than two right angles, the defect being a finite angle: these Lines are intersectional on that side of the transversal.

Let , make with the interior angles , together less than .

Make angle equal to angle ; produce to ; from draw at right angles to , and at right angles to .

Hence is also at right angles to ;

[II. Prop. 9.

i. e. , have a common perpendicular;

each is equidistant from the other;

[II. Prop. 7, Cor. 1.

also = the common distance between them.

[Pg 29]

Now angle is the defect, from , of the sum of the 2 interior angles , ;

hence, by hypothesis, it is finite;

it may be so multiplied, by a finite number, as to exceed angle .

[II. Ax. 2.

Call this finite number '.'

In , produced if necessary, take -times ; from draw at right angles to ; turn Triangle about into position ; then, about , into position , and so on, till there are such Triangles altogether; and let its final position be .

Then angle is -times angle ; i. e. it is greater than angle .

Let cut broken-Line at ; and join .

Then is greater than , which is greater than , which is greater than ;

[Euc. I. 20, 17, 19.

is greater than ;

but is the same fraction of which is of ;

is greater than ; i. e. the distance of , from , is greater than the common distance between and .

, are intersectional towards , .

Therefore, if a Pair of Lines &c.

Q.E.D.

[Pg 31]

APPENDICES

TO

PART I.

[Pg 32]

APPENDIX I.

Containing an alternative Axiom, which may be substituted for Axiom 1 at p. 14.

Definition.

The Segment, cut off, from any Sector of a Circle, by its Chord, may be called its 'outer Segment'; and the Triangle, contained by its Chord and its two Radii, may be called its 'central Triangle.' And, if its Arc be bisected and the point of bisection joined to the ends of its Chord, the isosceles Triangle, so formed, may be called its 'inscribed isosceles Triangle.'

PROPOSITIONS.

PROP. A. Theorem.

[Pg 33]

If, in any Sector of a Circle, its Chord be not-less than its Radius: then, in a Sector whose vertical angle is twice as great, its outer Segment is greater than its central Triangle.

Let be a Sector whose Chord is not-less than its Radius . Make angle equal to angle ; and join , .

Then vertical angle of Sector is twice as great as that of Sector .

It shall be proved that its outer Segment is greater than its central Triangle .

Because is not-less than ;

Triangle is not-less than Triangle ;

[I. Prop. 2.

, a fortiori, Segment is greater than Triangle .

Hence if, in any Sector &c.

Q.E.D.

PROP. B. Theorem.

[Pg 34]

If, in any Sector of a Circle, each of the equal Sides of its inscribed isosceles Triangle be not-less than its Radius; and if its outer Segment be greater than a certain multiple of its central Triangle: then, in a Sector, whose vertical angle is twice as great, its outer Segment is greater than twice that multiple of its central Triangle.

Let be a Sector such that each of the equal sides of its inscribed isosceles Triangle is not-less than its Radius , and such that its outer Segment is greater than times its central Triangle . Make angle equal to angle ; bisect angles , by Lines , ; and join , , , .

Then vertical angle of Sector is twice as great as that of Sector .

It shall be proved that its outer Segment is greater than times its central Triangle .

Because angle is greater than angle , and that angle is less than angle ;

angle is greater than angle ;

is greater than ;

[Euc. I. 25.

but is not-less than ;

is greater than ;

Triangle is greater than Triangle ;

[I. Prop. 2.

to each of these add Triangle ;

Figure is greater than twice Triangle .

Again, Segment is given to be greater than times Triangle ;

Segments , are together greater than times Figure ; i. e. are together greater than times Triangle ;

, a fortiori, Segment is greater than times Triangle .

Hence if, in any Sector &c.

Q.E.D.

Axiom.

[Pg 35]

[An alternative Axiom, to be substituted for Axiom 1, at p. 14, if the Reader feel any difficulty in granting that Axiom. In this case, certain portions of the foregoing Propositions will also need to be replaced by new matter, which is hereto appended.]

In every Circle, the inscribed equilateral tetragon, multiplied by ('' being a certain selected finite number), is greater than any one of the Segments which lie outside it.

Note.—The reader can assign to '' any finite value which he finds large enough to induce him to accept this Axiom. For example, if he be willing to grant that 1024 times the Tetragon is greater than the Segment, he can assign to it the value '10.'

PROP. C. Theorem.

[Pg 36]

[To be substituted for Prop. I, at p. 15.]

An isosceles Triangle, whose vertical angle is of a right angle, has its base less than either of its sides.

Let of a right angle be represented by '.'

Let be an isosceles Triangle, whose vertical angle at is .

It shall be proved that is less than .

Draw at right angles to , and, with centre , and distance , describe Quadrant. (See Note.) And join .

Then Triangle is one-fourth of an equilateral Tetragon inscribed in the Circle.

Hence times this Triangle is greater than Segment .

[Alternative Axiom.

Now, if we deny that is less than , we must assert that is not-less than .

Let this be our Hypothesis.

Note.—The Reader is requested to imagine a chord drawn to the arc .

[Pg 37]

Hence it may be proved, as in Book II, Prop. I, that any Chord, drawn from to any Point on the Arc , is greater than , and therefore not-less than .

Now, on our Hypothesis, is a Sector whose Chord is not-less than its Radius;

, in a Sector whose vertical angle is , its outer Segment is greater than its central Triangle;

[Prop. A.

i.e., in a Sector whose vertical angle is , each of the equal sides of its inscribed isosceles Triangle is not-less than its Radius, and its outer Segment is greater than once its central Triangle;

, in a Sector, whose vertical angle is , its outer Segment is greater than twice its central Triangle;

[Prop. B.

, similarly, in a Sector, whose vertical angle is , its outer Segment is greater than 4 times its central Triangle;

and so on;

, ultimately, in a Sector, whose vertical angle is , its outer Segment is greater than times its central Triangle;

but ;

, in Sector , Segment is greater than times Triangle .

But this is absurd, since it has been already proved less than times this Triangle.

Hence our Hypothesis, that is not-less than , is false; i. e. is less than .

Therefore an isosceles Triangle &c.

Q.E.D.

Corollary to Prop. C.

[Pg 38]

Hence, by Book I, Prop. III, it is possible to describe, on a given base, an isosceles Triangle having each base-angle equal to of a right angle.

[N.B. If the Reader be willing to grant, as axiomatic, that 1024 times the Tetragon is greater than the Segment, he must now admit, as logically proved, that an isosceles Triangle, whose vertical angle is of a right angle, has its base less than either of its sides. If such a Triangle were actually drawn, having each side 140 yards long, its base would be found to be less than an inch!]

PROP. D. Theorem.

[To be substituted for Prop. II, at p. 18.]

The angles of any Triangle are together not-less than of a right angle.

[The proof of Prop. II will serve here, without any change, except the substitution of '' for ''.]

[N.B. If the Reader be willing to grant, as axiomatic, that 1024 times the Tetragon is greater than the Segment, he must now admit, as logically proved, that the angles of any Triangle are together not-less than of a right angle.]

PROP. E. Theorem.

[Pg 39]

[To be substituted for Prop. III, at p. 19.]

There is a Triangle whose angles are together not-less than two right angles.

[The proof of Prop. III will serve here, without any change, except the substitution of '' for ',' and '' for 'one-eighth,' down to the words 'A similar proof &c.' at foot of p. 21; for which the following is to be substituted.]

A similar proof will hold for , , and so on; and ultimately for .

Hence, either there is an 'amount' not-less than , or else every 'amount' is not-less than .

But .

Hence the second clause of this alternative contains the first.

Hence the first clause must be true.

That is, there is a Triangle &c.

Q.E.D.

APPENDIX II.

Is Euclid's Axiom true?

§ 1.

[Pg 40]

Infinite and Finite Magnitudes.

The answer I propose to give to this alarming question is that, though true for Finite Magnitudes—the sense in which, as I believe, Euclid meant it to be taken—it is not universally true.

Will the gentle Reader be so kind as to join me in contemplating, for a few minutes, the Infinite Space which surrounds our tiny planet? We believe—those of us, at least, who answer fully to the ancient definition of Man, 'animal rationale'—that it is infinite. And that, not because we profess to have grasped the conception of Infinity, but because the contrary hypothesis contradicts Reason: and what contradicts Reason we feel ourselves authorised to deny. Both conceptions—that Space has a limit, and that it has none—are beyond our Reason: but the former is also against our Reason: for we may fairly say "When we have reached the limit, what then? What do we come to? There must be either Something, or Nothing. If Something, it is full Space, 'plenum': if Nothing, it is empty Space, 'vacuum.' That there should be neither of these is a logical impossibility. Such an hypothesis would be—in the[Pg 41] words of Master Constable Dogberry—'most tolerable and not to be endured.'"

I propose to show, by certain considerations which begin with Infinite Space, but will speedily condescend to Finite Magnitudes, that it is possible for two homogeneous Magnitudes to be so related to each other that no multiple of the lesser will exceed the greater. (It is of course assumed that a 'multiple' of a Magnitude is the

result produced by the use of a 'multiplier,' and that a 'multiplier' is a nameable—and therefore a finite—number.)

"Yet surely," the gentle Reader will protest, "Euclid has assumed the exact contrary of this? Does he not, in Book X, Prop. 1, tacitly assume the Axiom that the lesser of two Magnitudes may be so multiplied as to exceed the greater?"

Gentle Reader, he does! But my contention is that, in so doing, he excludes from his view both Infinities and Infinitesimals, and is contemplating Finite Magnitudes only.

For consider Euclid's Definitions of the word 'Ratio' and of the phrase 'to have a Ratio to.' (Book V. Def. 3, 4.)

(3) λόγος ἐστὶ δύο μεγεθῶν ὁμογενῶν ἡ κατὰ πηλικότητα πρὸς ἄλληλα ποιὰ σχέσις.

(4) λόγον ἔχειν πρὸς ἄλληλα μεγέθη λέγεται, ἃ δύναται πολλαπλασιαζόμενα ἀλλήλων ὑπερέχειν.

Quite literally, these are:—

(3) "Ratio is a certain relationship, as to size, of two homogeneous Magnitudes, each to the other."

(4) "Magnitudes, which can, (on) being multiplied, exceed each the other, are said to have a Ratio, each to the other."

But they become more intelligible when less literally translated:—

(3) "Ratio is a certain relationship, as to size, borne, by a Magnitude, to another Magnitude homogeneous with it."

[Pg 42]

(4) "A Magnitude is said 'to bear a Ratio to' another Magnitude, homogeneous with it, when either of them, that is not greater than the other, can be made so by multiplication."

Some translators introduce the word 'mutual' into No. (3), and tell us that Ratio is 'a mutual relation of two Magnitudes': but this seems to me incorrect, as seeming to imply that the Ratio, borne by to , is identical with that borne by to . But, if were 3-4ths of , would not be 3-4ths, but 4-3ds, of : and the Ratio '4-3ds,' though of the same nature as the Ratio '3-4ths,' is not identical with it.

Now it seems to me clear that Euclid does not mean to imply that any two homogeneous Magnitudes bear 'Ratios' to each other: for in No. (4) he gives us a test, by which to know in what cases two such Magnitudes do, and in what cases they do not, bear 'Ratios' to each other. This test would be wholly superfluous if it were true, of any two homogeneous Magnitudes, that one could be said 'to bear a Ratio to' the other.

What cases then, does Euclid mean to exclude by this test? My answer is "all cases in which one of the Magnitudes is infinitely greater than the other." Take, as an

example, these two Magnitudes—a Cubic Inch, and Infinite Space. It is not possible, by multiplying a Cubic Inch by any finite number, however great, to make it exceed, or even equal, Infinite Space: hence Euclid's test fails in this case, and Euclid would, no doubt, decline to say that either of these Magnitudes, though they are strictly homogeneous, bears a 'Ratio' to the other.

My conclusion, then, is that, in Book X. Prop. 1, Euclid is limiting his view to the case of two homogeneous Magnitudes which are such that neither of them is infinitely greater than the other: nay, more—for such a limitation would not exclude the case of two Infinities of the same order—that he is contemplating Finite Magnitudes only.

§ 2.

[Pg 43]

Infinitesimal Lines and Strips.

We have already seen that, in the case of two homogeneous Magnitudes, one Finite and the other Infinite, no multiple of the lesser will exceed the greater: and I now propose to show that it is possible for the same thing to happen in the case of two homogeneous Magnitudes, neither of them being Infinite.

Let us imagine an Infinite Plane, placed upright, and facing us—as if we were standing in front of a wall, which extended to infinity, upwards, downwards, to the right-hand, and to the left. If we divide this Plane by a single horizontal straight Line, extended to infinity to the right-hand and to the left, we get half of the whole Plane above the Line, and half below it, wherever we choose to place the Line. This may be deduced from the logical principle that we have no reason for believing either to be greater than the other; or we may adopt M. Bertrand's Axiom, "Two spaces, whether finite or infinite, are equal, when one can be placed upon the other so that any point whatsoever of either coincides with a point of the other"—a condition which seems, theoretically, readily attainable, by making one portion of the Plane revolve round the boundary-line, as a hinge, till it coincides with the other portion. Each portion is, of course, an Infinity of the same order as the whole Plane.

Now let us imagine two such infinite horizontal straight Lines, 'separational' from each other (i. e. never intersecting), placed at a finite distance apart, and therefore having between them a Strip, finite in width, infinite in length, and therefore infinite in area.

Now it clearly is not possible, by multiplying this Infinite Strip by any finite number, however great, to make it exceed, or even equal, the whole Infinite-Plane. Here again, then, Euclid's test fails, and neither of these Magnitudes, though they are homogeneous can be said to have a 'Ratio' to the other.[Pg 44] In fact, the Infinite-Strip is an Infinity of a lower order than the Infinite-Plane.

Comparing this Strip with a single square-inch, you will, I fancy, be willing to grant at once that no multiple of the latter can possibly reach—much less exceed—the former. We have, in fact, established the existence of three kinds of Area, viz. Finite Areas (e.g. a square inch), Infinities of the first order (e.g. the Infinite-Strip we have

been considering), and Infinities of the second order (e.g. the upper half of the whole Infinite-Plane).

Now let us go a step further. Let the two sides of this Strip be supposed to gradually approach each other—still maintaining their 'separational' character—and let us consider the effect of this process on the intervening area.

When the width of the Strip has been reduced to half-an-inch, you will grant, I suppose, that the area is exactly half what it was at first—since two such Strips, laid side by side, evidently make up the original Strip. And so, by reducing the width to a quarter-inch, &c., we obtain a number of different areas, all Infinite alike, and yet having finite ratios to one another: in fact, so long as the width continues to be a finite fraction (i. e. a fraction with a finite numerator and denominator) of an inch, the area continues to be an Infinity of the same order as the original Strip. (It seems obvious that Infinities of the same order have finite ratios to one another, and that any one of them can be so multiplied, by a finite number, as to exceed any other.)

But this narrowing process may be continued until the two Lines absolutely coincide: and what then becomes of the area? It cannot be denied that it is then Zero.

Now I have ventured (see p. 2) to lay it down, as an Axiom, that, when a Magnitude varies continuously, and has changed from a certain value to a certain other value, it must have passed through every intermediate value. And what intermediate values do we find between an Infinite Area and Zero?[Pg 45] Surely every Finite Area, that can be named, lies between them? The Reader can, if he likes, part company with me at this point: but to my Reason it seems absolutely clear—first, that the Strip does diminish continuously, and not 'per saltum'; and secondly, that its area has, at some time or other during the process, every conceivable finite value. At one time, for instance, it contains a square-mile: and, rather later in its career, it is reduced to a single square-inch.

Let us contemplate it in this last-named condition. Its length? As Infinite, clearly, as it was at first. Its Area? One square-inch, undoubtedly. And what is its width?

Can you, oh gentle Reader, find any reasonable answer to this question, except the following? "Its width is infinitely small—having, in fact, exactly the same relation to a linear inch which that linear inch has to an infinite Line."

If this be so (and I see no way out of it), we have found two Magnitudes, both linear, neither of them infinite, and yet such that no finite multiple of the lesser can possibly exceed the greater. They are, in fact, not of the same order; one of them being Finite, and the other an Infinitesimal of the first order.

But is not the gentle Reader saying to himself, all this time, "I cannot believe in the existence of a Strip, Infinite in length, and yet Finite in Area! No doubt, if such a Strip could exist, its width must be what you call 'Infinitesimal'; since a Finite width must give an Infinite area. But I don't believe in the existence of an Infinitesimal width! My belief is that, if you make the edges of the Infinite-Strip gradually approach till they coincide, its width will continue Finite till the last moment, and will then suddenly become Zero; and that its area will continue an Infinity of the first order till the last

moment, and will then suddenly become Zero."

[Pg 46]

Very good. My gentle Reader has formulated his views, very clearly and definitely. Permit me now to offer to his consideration a Strip, which I will prove to be at once Infinite in length and Finite in area.

Let , , be rectangular Axes of Reference; and let us trace the Curve , where = unit-line. Hence, when , ; when , ; when , ; when , ; and so on. Hence the Curve is . Also, however large becomes, can never be Zero; i. e. the Strip, intercepted between the Curve and the -Axis, and bounded at the left-hand end by , is Infinite in length. And now let us estimate its area. Its first portion, , is less than the rectangle ; its second portion is less than ; and so on. Hence its area is less than ; i. e., is less than + &c. for ever; therefore, a fortiori, it is less than 2.

Now an area, which is less than '2' is surely Finite? Does the gentle Reader see any escape from admitting this? And, if he admits this, does he still maintain that the width of this Infinite-Strip (which obviously dwindles, as you go along the Strip, but never becomes Zero) never ceases to be Finite? Yet surely a Strip, Infinite in length, and nowhere less than Finite in width, must be Infinite in area?

In brief, I place before my gentle Reader that savouriest of Logical dishes, a Trilemma! Either he must assert that a Strip, Infinite in length, and nowhere less than Finite in width, is only Finite in area; or he must assert that the length of this Strip is Finite, i.e. he must assert that the Curve meets the x-Axis; or else he must admit that its width ceases to be Finite[Pg 47] without becoming Zero, i.e. he must admit that its width becomes Infinitesimal! Let him take his choice, and help himself. 'May good digestion wait on appetite, And health on both!'

Now let us cut off, from the infinitely-long Strip named in p. 45, whose area is a square-inch, a piece just an inch long. What will its area be? It is evident that no finite multiple of this short Strip can ever make up the infinitely-long Strip; that is, no finite multiple of its area can make up a square-inch. Hence its area must be an Infinitesimal of the first order.

But this Infinitesimal area, inconceivably small as it is, is nevertheless greater than Zero. Hence our continuously-diminishing Strip is bound, before reaching Zero, to pass through this singularly unassuming value. And, at that moment, what will be its width? Its length will be Infinite, as usual: its area will be an Infinitesimal of the first order: but its width cannot be an Infinitesimal of the same order as the previous width; for that would yield a finite area, as we have seen. What else, then, can it be but an Infinitesimal of the second order? A Line of such stupendous brevity that no finite multiple of it can even make up an Infinitesimal of the first order. Evidently we might repeat this process ad libitum, and so get Infinitesimals of the third order, the fourth order, and so on.

To sum up our results, so far. We see that a Line may be either Finite, or may extend to an Infinity of the first order, or may dwindle to an Infinitesimal of the first, second, or any order we choose: and that an Area may be either Finite, or may extend

to an Infinity of the first order (an Infinite-Strip), or of the second order (e.g. the whole Infinite-Plane), or may dwindle to an Infinitesimal of any order we choose.

Now we may reasonably expect to find that all, that has been here said, is equally applicable to any kind of Magnitude that is capable of continuous increase and decrease. Let us consider, then, whether it is possible to have Infinitesimal Angles of the various orders.

§ 3.

[Pg 48]

Infinitesimal Angles and Sectors.

For this purpose, let us return to our upright Infinite-Plane, and, taking some Point at random as a centre, let us imagine two Lines radiating from it (say at an angle of 45°), and both of them extended to infinity, and therefore having between them a Sector, finite in angular magnitude, infinite in length, and therefore infinite in area.

I named 45° as my specimen-angle, because it is the one single angle, other than a right angle, with which Society is acquainted. Enquire of some chatty traveller, who is relating his experiences of an Alpine Pass, what slope it was that he had to climb. "The ground sloped at an angle of forty-five," he is sure to reply. Nay, I once met a gentleman who, on hearing it mentioned that the 'dip' of a certain river-bed was, in one place, "one in forty-five," cautiously remarked "I suppose that means an angle of forty-five degrees?" Imagine a river sloping at that angle! And then imagine the labour of rowing up it, and the headlong, wild delight of rowing down it! But this is a digression.

Now it clearly is possible, this time, by multiplying this Infinite-Sector by a certain finite number, namely '8,' to make it equal to the whole Infinite-Plane. Hence this Infinite-Sector is an Infinity of the same order as the whole Infinite-Plane; i.e. it is of the second order.

Now let us suppose that the two sides of this Infinite Sector gradually approach each other until they coincide. It cannot be denied that the area then becomes Zero. It has dwindled then, from an Infinity of the second order (when its angular magnitude was finite), down to absolute Zero. And there seems no room to doubt that it has done this continuously, and not 'per saltum.' What values, then, has it gone through on the way? Can we reasonably doubt that it has gone through, first, all Infinities of the first order (during which process its angular magnitude would be an Infinitesimal of the first order), secondly, all Finite values (its angular magnitude being an Infinitesimal of the second order), thirdly, all Infinitesimals of the first order (its angular magnitude being an Infinitesimal of the third order), and so on—the angular magnitude being always two degrees ahead of the area, in this long and fatiguing competition in the Dwindling-Race.

§ 4.

[Pg 49]

Pairs of Lines.

We are now in a position to examine the phenomena of intersection, or non-intersection, with regard to a Pair of Lines, by imagining one of them to revolve about a fixed Point.

Let be one of the 2 Lines: and let be, at first at right angles to it: and let it then revolve, about , so as to take the successive positions , , , its final position being at right angles to its first position, and therefore parallel to .

Let us further suppose that the angle, contained between and the upper part of the revolving Line, dwindles, from a right angle, through all possible finite lesser values, while its[Pg 50] upper edge revolves from , to ; that, the moment the upper edge goes below , this angle becomes an Infinitesimal of the first degree, and so dwindles, through all such values, while its upper edge revolves from to ; that, the moment the upper edge goes below , this angle becomes an Infinitesimal of the second degree; and so on.

Now let us suppose all the Lines in the diagram to extend to infinity both ways: and let us call the Infinite-Sector, lying between -produced and the upper part of the revolving-Line, 'No. 1'; the semi-Infinite-Strip (I mean, by 'semi-Infinite,' that it is terminated at one end), lying between the Lines -produced and -produced, and bounded at the right-hand end by , 'No. 2'; the surface, lying between the Line -produced and the lower part of the revolving-Line, and bounded at the left-hand end by , (which will be a Triangle, so long as -produced and the lower part of the revolving-Line continue to intersect, and will become a semi-Infinite-Strip when they cease to intersect,) 'No. 3'; and, with regard to the Infinite-Sector lying between the Line -produced and the lower part of the revolving-Line, let us call that portion of it, which lies above -produced (which portion will be a semi-Infinite-Strip, so [Pg 51]long as -produced and the lower part of the revolving-Line continue to intersect, and will become the whole Infinite-Sector if they should cease to intersect before the revolving-Line reaches the position), 'No. 4'; and that portion of it, which lies below -produced (which portion will cease to exist as soon as these Lines cease to intersect) 'No. 5.'

Let us now investigate the changes, in the areas of these 5 surfaces, caused by the changes in the position of the revolving-Line.

First as to No. 1. This is clearly, throughout its history, an Infinite-Sector, whose vertical-angle is at first a right angle, and ultimately Zero. Also its area is at first one-quarter of the whole Infinite-Plane, and ultimately Zero. Also, so long as the upper part of the revolving-Line ranges between and , the area continues to be an Infinity of the second order; and, as the revolving-Line crosses the position , the area changes, from a very small (!) Infinity of the second order, to a very large Infinity of the first order. Similarly, as the revolving-Line crosses the position , the area changes, from a very small Infinity of the first order, to a very large Finite value.

A little further on, it will of course become an Infinitesimal of the first order; and so on, through the other orders, till it finally reaches the value Zero.

Next, as to No. 2. This is clearly, throughout its history, one-half of the Infinite-Strip lying between and , and is therefore an Infinity of the first order. Its

area is a constant quantity, being unaffected by the revolving-Line.

Next, as to Nos. 4, 5. (I take these next, because they will help us to investigate the properties of No. 3.) It is evident that, so long as No. 5 continues to exist, the two together constitute—and that, when No. 5 has ceased to exist, No. 4 by itself constitutes—an Infinite-Sector, which is the exact counterpart of No. 1—the two having what Euclid calls 'opposite vertical angles.' Hence, while the lower part of the revolving-Line ranges between and [Pg 52], the area of No. 4 continues to be an Infinity of the second order, and entirely declines to be 'cribb'd, cabin'd, and confined' within such narrow limits as our Infinite-Strip! Hence the revolving-Line continues, all this time, to intersect -produced. (N.B. Here we have a proof of the truth of Euclid's Axiom, when amended, as I have done at p. 28, by inserting the words 'the defect being a finite angle.')

Let us now, reserving for future consideration the phenomena of the period while the upper part of the revolving-Line is crossing from to , suppose it to have passed , so that the angle, which it makes with , has become an Infinitesimal of the second order. Will its lower part continue to intersect -produced? It may easily be shown, by a reductio ad absurdum, that it will not: for, if it did, Nos. 1, 2, 3 would then make up an Infinite Sector, whose vertical angle would be an Infinitesimal of the second order, and whose area would therefore be finite. But the area of No. 2 is always an Infinity of the first order. Which is absurd. Hence, after the upper part of the revolving-Line has passed , its lower part does not intersect -produced.

We have now to answer a far more puzzling question, namely, what happens while the upper part of the revolving-Line is crossing from to ? Does its lower part intersect [Pg 53]-produced, all the while? Or does it fall clear of it, all the while? Or does it at first intersect it, and afterwards cease to do so?

First, suppose it to intersect -produced. In this case No. 3 is clearly a closed Triangle, whose vertical-angle is an Infinitesimal of the first order. This looks as if its area must also be an Infinitesimal of the first order: but this, we know, cannot be, since it contains within it the finite area possessed by No. 3 while the upper part of the revolving-Line was passing from to . Hence its area must be, at least, finite. The only way I can see out of this difficulty is to assume that the sides of this Triangle have become infinite; i.e. that the revolving-Line intersects -produced at an infinite distance.

Next, suppose it to fall clear of -produced. In this case No. 3 is a semi-Infinite-Strip, but we cannot be certain that its area is, like such Strips when of a uniform width, infinite; for its width dwindles so much towards the right-hand that it may possibly be, in the early part of the period, finite. I see no way of settling this question; but, luckily, it is not relevant to the question of intersection.

My own inclination is to believe that, during this second period of revolution, the lower part of the revolving-Line at first intersects -produced, at an infinite distance, and then ceases to intersect it.

After the revolving-Line has once ceased to intersect -produced, No. 4 is of course equal to No. 1. That is to say, the surface, lying between the whole revolving-Line and

, is, from that moment until the revolving-Line coincides with , of a constant area, since it is the sum-total of Nos. 1, 2, 3, which is equal to the sum-total of Nos. 2, 3, 4, i.e. to the whole Infinite-Strip. And this will continue true, while No. 1 and No. 4 dwindle down, through all finite and infinitesimal values, till they finally reach Zero.

[Pg 54]

The results we have arrived at will perhaps be more easily understood by examining the following Table of values. The symbols used in it are as follows:—

Symbols. Meanings.

a right angle.

a large finite magnitude

a small " "

a large Infinitesimal of 1st order.

" " 2nd "

a large Infinity of 1st order.

area of the semi-Infinite-Strip No. 2.

[an Infinity of 1st order.]

area of one-fourth of the Infinite-Plane.

[an Infinity of 2nd order.]

Position of
upper part
of revolving-
Line. Vertical
angle of
Space
No. 1. Areas of Spaces.
No. 1. No. 2. No. 3. No. 4. No. 5.
Zero
"
"
"
"
" ? ? ?
"

"

does not

" "

exist

" "

"

" "

"

&c. &c. " " &c. "

Zero. Zero. " " Zero. "

[Pg 55]

The sum of the whole matter appears to be this. If a Pair of Lines make, with a certain transversal, two interior angles on the same side of it together less than two right angles, then, so long as the defect is finite, there is no doubt that the Lines intersect: also, if the theory be true, that the area of an Infinite-Sector, whose vertical-angle is finite, and whose area is therefore undoubtedly an Infinity of the second order, passes, as its vertical angle dwindles to Zero, through infinite values of the first order, and then through finite values, its vertical angle meanwhile passing through infinitesimal values of the first and second order—if all this be true, it follows that, when the 'defect from two right angles' becomes an Infinitesimal of the first order, the Lines may possibly intersect, but can only do so at an infinite distance; and that, when the defect has become an Infinitesimal of the second order, the Lines have ceased to intersect.

The theory, here discussed, may be bewildering; but it is at least consistent with itself: and it seems to me to be quite as credible as the theory that 'Infinitesimals' are mythical, and that a Finite Magnitude, dwindling down to Zero, continues Finite to its last gasp.

[Pg 56]

Another process—a negative one—has occurred to me, for disproving the absolute truth of Euclid's Axiom. It is a very simple process, and has the great recommendation of not requiring any belief in the existence of Infinitesimals.

On an upright Infinite-Plane let us suppose 2 horizontal Infinite-Lines, having a common perpendicular, and therefore of course never intersecting. Let us call them 'No. 1' and 'No. 2,' No. 1 being above No. 2: and let us suppose the common perpendicular to be an inch long, so that the 2 Lines are the edges of an Infinite-Strip, whose uniform width is an inch.

Now let us suppose Line No. 1 to begin to revolve about the upper end of the common perpendicular. The believer in the absolute truth of Euclid's Axiom is bound to believe that, in the very act of beginning to revolve, it also begins to intersect No.

2: he cannot allow the one process any such start of the other as might enable him to say "No. 1 has begun to revolve, but has not yet begun to intersect No. 2": such a state of things must be, in his view, absolutely impossible. 'And what's impossible ca'n't be, And never never comes to pass!'

Very good. The actual process of beginning to intersect No. 2 is a deep mystery, no doubt: but that the thing happens is quite undeniable. We are able to say "Now it isn't intersecting No. 2—and now it is!" So, though we cannot conceive how it managed to begin, it has certainly done it.

Now let us suppose another horizontal Line, 'No. 3,' lying an inch below No. 2. The believer in Euclid's Axiom is bound to assert, as to No. 3, exactly what he asserted as to No. 2. Has he, then, any logical escape from the conclusion that the revolving Line begins to intersect Nos. 2 and 3 together? I see none, myself. And yet how can it get at No. 3, without first going through No. 2? Any point on No. 1 (I am careful not to say 'every': I know how gleefully the logical Reader would swoop down upon me with the crushing sarcasm "What does 'every' mean, when there is no limit to the number of points?": but 'any' is a safe epithet) is amenable to the simple rule of[Pg 57] "First come, first served. Cross No. 2 first: and afterwards (not by any means simultaneously) you have our gracious permission to cross No. 3." Now, what is true of any point on the Infinite-Line No. 1 is surely true of that Infinite-Line itself? To say "No. 1 intersects No. 3" is tantamount to saying "A certain point of No. 1 has reached No. 3." And how did that Point get below No. 2?

Of two things, one. Either some point of No. 1 has crossed Nos. 2 and 3 at the same moment: or else no point of No. 1 has crossed No. 3 until after it had crossed No. 2. That is a logical Dilemma. Which of its two horns does the Reader prefer?

The choice of the first horn involves the Reader's acceptance of ubiquitous points! In the event of his choosing the second horn, he seems logically bound to admit it as a possible state of things, that No. 1 should have begun to revolve, and yet should not have begun to intersect No. 3—which is a surrender of his belief in the universal truth of Euclid's Axiom.

My final answer, then, to the question "Is Euclid's Axiom true?" is as follows:—

"If the defect, from the sum of two right angles, be finite, the Lines certainly meet: if it be an Infinitesimal of the first order, they may meet, or not, according to circumstances: if it be an Infinitesimal of the second, third, or any higher order, they certainly do not meet."

The question, with which this Appendix is headed, was sufficiently startling: but a more startling one remains to be answered. "If Euclid's Axiom be not universally true, what becomes of all the Propositions which he has made to depend upon it, such as I. 29? Has he done no more than prove them to be partially true?"

To this disheartening question I will give as re-assuring an answer as the case seems to admit of. It must be admitted that Euc. I. 29 requires—if we would make it strictly true—the same limitation as the Axiom: it should run as follows:—"Two[Pg 58] Lines, which do not meet, make, with all transversals, angles which are equal so far

as finite differences are concerned" (i. e. angles so nearly equal that the difference, if any, is infinitesimal). And of course Euc. I. 32, as proved from Euc. 1. 29, would need a similar qualification. It seems to me very doubtful whether Euclid ever noticed this defect in his Axiom. If he did, it is possible that he may have thought fit to ignore it, on the ground that, when Finite Magnitudes differ only by an Infinitesimal, they are, for all practical purposes, equal.

But I feel bound to admit that, for the purpose of proving Euc. I. 32 to be, as it really is, universally true, neither Euclid's Axiom, nor any other that deals with intersection of Lines, will suffice: and that some Axiom, not involving that principle, must be substituted for it.

Let me say in conclusion that, though I assert the absolute truth of Euclid's Axiom—with the limiting clause I have introduced, 'the defect being a finite angle'—it still remains, in my opinion, a 'disputable' Axiom; i.e. it is not properly admissible as an Axiom, but ought to be, if possible, proved as a Theorem.

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APPENDIX III.

How should Parallels be defined?

We know that, if a Pair of Lines has either of the following properties

(1) they are equally inclined to a certain transversal,

(2) one of them contains 2 Points on the same side of, and equidistant from, the other,

it has all the following properties

(3) they are equally inclined to all transversals,

(4) any 2 Points, on either of them, are on the same side of, and equidistant from, the other,

(5) they do not meet, however far produced.

Any one of these properties may be used as a Definition. Let us take them one by one.

No. (1). This has the advantage that we need not begin by proving that such Lines exist, but may assume it as axiomatic. It has been used as a Definition by Varignon, Bezout, Cooley, &c.: at least, they say "which make equal angles with[Pg 60] a transversal," leaving it uncertain whether they mean "a certain transversal" or "any transversal": if the latter, it is of course No. (3) they are proposing to use. From No. (1) we can prove No. (5) without using any disputable Axiom (see Euc. I. 27, 28), but not No. (3) or No. (4).

No. (2). This has the same advantage as No. (1). It has been used as a Definition by D'Alembert. From it, also, we can prove No. (5) without using any disputable Axiom; and it, also, fails to prove No. (3) or No. (4).

No. (3). This cannot be used, as a Definition, till we have proved that such Lines exist—which has not yet been done without employing some disputable Axiom. If Varignon, &c. mean this to be their Definition, they are assuming the existence of such Lines,—a very un-axiomatic Axiom. But, when once their existence has been granted, or proved, No. (4) can be deduced.

No. (4). This cannot be used, as a Definition, till we have proved that such Lines exist—which has not yet been done without employing some disputable axiom. When once their existence has been granted, or proved, No. (3) can be deduced. No. (4) has been used as a Definition by Wolf, Boscovich, T. Simpson, and Bonnycastle.

No. (5). This has the advantage that it is easy to prove (as in Euc. I. 27, 28) that such Lines exist. It has been used as a Definition by Euclid and a host of other geometers. It has, however, the enormous disadvantage that, whereas Nos. (3), (4), give us a unique Pair of Lines (e.g. given a Line and a Point not on it, we can prove that only one Line can be drawn, through the Point, such that the Pair shall have property No. (3)), No. (5) does not: on the contrary, given a Line and a Point not on it, a whole 'pencil' of Lines may be drawn, through the Point, and not meeting the given Line: all we need to do is to take care, after drawing one such Line, that the others shall make with it angles which are Infinitesimals of the second order with regard to a right angle.

[Pg 61]

We see, then, that the word "Parallels" has been already used with four, and possibly with five, different meanings: so that any fresh writer, who uses the word, is liable to be misunderstood unless he first defines it. The derivation of the word would seem to suggest No. (4) as its Definition; but Euclid's adoption of No. (5) has led to that being the popular meaning attached to the word.

It is easy, however, to avoid all this ambiguity by the use of new terms. Rejecting Nos. (1) and (2), as useless for the purpose of definition, we may call a Pair of Lines, which possesses

No. (3), "equiangulated";

No. (4), "equidistantial";

No. (5), "separational";

and thus avoid the dangerous word "parallel" altogether.

My own belief is that No. (3) is, on the whole, the property best adapted for practical use as a Definition.

APPENDIX IV.

How the Question stands to-day.

I will first enumerate certain Theorems connected with Pairs of Lines.

I will then notice three other methods which have been suggested for superseding Euclid's Axiom.

And, in conclusion, I will indicate what seem to me the most hopeful directions for future efforts at exploring this fascinating, but very obscure, region of mathematical research.

§ 1.

[Pg 62]

Certain universally-true Theorems, provable from genuine Axioms (i. e. from Axioms whose self-evident character is indisputable).

(1) A Pair of Lines, which are equally inclined to a certain transversal, are non-intersectional (i. e. never intersect, however far produced). [Euc. I. 27, 28.]

(2) [contranominal of (1)] A Pair of intersectional Lines (i. e. which either intersect or would do so if produced) are unequally inclined to any transversal. [Euc. I. 16, 17.]

(3) A Pair of Lines, such that two Points on one of them are on the same side of, and equidistant from, the other, are non-intersectional.

(4) [contranominal of (3)] A Pair of intersectional Lines are non-equidistantial (i. e. are such that any two Points on either of them, which are on the same side of the other, are non-equidistant from it, that which is further from the Point of intersection being also further from the other Line).

(5) If there be a Triangle whose angles are together equal to two right angles: the angles of any Triangle are together equal to two right angles.

(6) There is a Triangle whose angles are together not-greater than two right angles.

§ 2.

[Pg 63]

Certain universally-true Theorems, not provable from genuine Axioms, but provable if any one of them be accepted as an Axiom.

[N.B. These will be hereafter referred to as the 'Nine Quasi-Axioms,' their self-evident character being disputable.]

(1) Through a given Point, outside a given Line, a Line may be drawn, such that the Pair shall be equally inclined to any transversal.

(2) A Pair of Lines, which are equally inclined to a certain transversal, are so to any transversal. [Deducible from Euc. I. 27, 28, 29.]

(3) [contranominal of (2)] A pair of Lines, which are unequally inclined to a certain transversal, are so to any transversal.

[Pg 64]

(4) If a Point move so as to be at a constant distance from a given Line, its path shall be a straight Line.

*(5) Through a given Point, outside a given Line, a Line may be drawn equidistantial from the given Line (i. e. such that any two Points on it shall be equidistant from the given Line).

(6) A Pair of Lines, such that two Points on one of them are on the same side of, and equidistant from, the other, are equidistantial (i. e. are such that any two Points on either of them are equidistant from the other).

(7) [contranominal of (6)] A Pair of Lines, such that two Points on one of them are non-equidistant from the other, are non-equidistantial (i. e. are such that any two Points on either of them, which are on the same side of the other, are non-equidistant from the other).

*(8) A Line cannot recede from and then approach another; nor can it approach and then recede from another while on the same side of it.

*(9) In any Circle, the inscribed equilateral Tetragon is greater than any one of the Segments which lie outside it.

If any one of these 9 Theorems be granted as an Axiom, the rest can be proved from it. But only 3 of them, so far as I know, have been used as Axioms—No. 5 by Clavius, No. 8 by Dr. R. Simson, and No. 9 by myself. Clavius' Axiom requires us to assure ourselves that it will continue true when the Lines are produced without limit; and the strain on the imagination, caused by the effort of following them into Infinite Space, is one to be, if possible, avoided. Dr. R. Simson's Axiom gains a certain plausibility from the fact that, for intersecting Lines, it admits of actual proof (see § 1. (4)): where, however, the Lines are not known to intersect, it is an appeal to the eye, of very doubtful force.

§ 3.

[Pg 65]

Certain universally-true Theorems, not provable from genuine Axioms, but provable if any one of the 'Nine Quasi-Axioms,' given in § 2, be accepted.

(1) There is a finite angular magnitude such that the angles of any Triangle are together not-less than it.

(2) There is a Triangle whose angles are together not-less than two right angles.

(3) There is a Triangle whose angles are together equal to two right angles.

(4) The angles of any Triangle are together equal to two right angles. [Euc. I. 32.]

§ 4.

Certain partially-true Theorems, not provable from any universally-true Axioms, whether genuine or 'quasi,' but provable if any one of themselves be accepted as an Axiom.

[N.B. By 'partially-true' is meant 'true for finite magnitudes.' They become universally-true, if 'magnitude' be taken to mean 'finite magnitude'; 'equal' to mean 'not differing by a finite difference'; 'unequal' to mean 'differing by a finite difference'; 'multiplied' to mean 'multiplied by a finite number'; and 'intersectional' to mean 'intersectional at a finite angle.'

These Theorems will be hereafter referred to as the 'Nine Pseudo-Axioms,' the name being chosen to indicate that they are not even universally true—far less self-evident.]

[Pg 66]

(1) The lesser of two homogeneous Magnitudes may be so multiplied as to exceed the greater.

(2) A Pair of Lines, which are unequally inclined to a certain transversal, are intersectional. [Euclid's Axiom.]

(3) [contranominal of (2)] A Pair of non-intersectional Lines are equally inclined to any transversal. [Euc. I. 29.]

(4) A Pair of Lines, such that two Points on one of them are non-equidistant from the other, are intersectional.

(5) [contranominal of (4)] A Pair of non-intersectional Lines are equidistantial.

(6) On either of two intersectional Lines a Point may be found, whose distance from the other Line shall exceed any assigned length.

(7) On either of two intersectional Lines a Point may be found, such that the distance, from its projection on the other Line to the point of intersection of the two Lines, shall exceed any assigned length.

(8) A Pair of intersectional Lines cannot be, both of them, non-intersectional with a third Line.

(9) [contranominal of (8)] A Pair of Lines, which are, both of them, non-intersectional with a third Line, are non-intersectional.

Six of these have been used as Axioms, viz. (2), which is Euclid's celebrated Axiom; (4), by T. Simpson; (6), by Proclus; (7), by Franceschini; (8), by Ludlam, Playfair, &c.; (1), by Legendre, in the 12th Volume of the 'Memoirs of the Institute,' being his "latest attempt" (I quote from De Morgan's article on Parallels in Knight's Cyclopædia) "at the solution of the problem." He only succeeds, however, in proving Prop. 6 at p. 10 of this book, and in proving that Prop. 8, at p. 26, follows logically from Prop. 4, at p. 22. In order to prove Euc. I. 32, he introduces the principle of Limits and Vanishing Quantities, which takes us at once into the region of Infinities and Infinitesimals. But a greater success than this has, I understand, rewarded some recent investigations made by Professor J. Cook Wilson, of Oriel College, Oxford, who has deduced Euclid's 12th Axiom from No. (1) of this Section.

In all these systems, however, including Euclid's, the deductions, from the proposed Axiom, labour under the same defect as the Axiom itself, that is, they are

only partially, and not universally, true.

§ 5.

[Pg 67]

Other methods of treatment.

Three other methods of treating this subject call for notice.

Playfair and (more recently) Wilson have tried to deduce the properties of Parallels (and thence Euc. I. 32) from the idea of sameness of direction, as predicated of two non-coincidental Lines (i. e. which do not lie in one and the same straight Line).

The foundation-stones of this Theory—without which it has no raison d'être whatever—are the two Axioms, which must necessarily be somewhere assumed, whether expressly or tacitly, first, that it is possible for non-coincident Lines to have 'the same direction'; and secondly, that Lines, which have the same direction, make equal angles with all transversals.

Before discussing the first of these two Axioms, permit me to remind the Reader that the question before us is not "is it true?" but "is it axiomatic?", that is, 'does the average human intellect accept it as true, without proof?' It is a question in Mental Physiology rather than in Geometry. The 47th Proposition of Euclid is quite as true as the Axiom 'things that are equal to the same are equal to one another'; but the average human intellect, while accepting the latter without proof, does most certainly demand a good deal of proof before it will accept the former. Intellectual beings may conceivably exist, to whom Euc. I. 47 is axiomatic; but our books are not written for them.

Now there is one preliminary step, that is absolutely indispensable before the human intellect can accept any Axiom[Pg 68] whatever: and that is, it must attach some meaning to it. We cannot, rationally, either assent to, or deny, any Proposition the words of which convey to us no idea.

We have, then, two questions to answer: first, "what idea is conveyed to the average human intellect by the phrase 'in the same direction,' when applied to non-coincident Lines?", secondly, "in accepting the Axiom, that 'Lines, which have the same direction, make equal angles with all transversals,' to what other assertions are we committing ourselves?"

If we contemplate a fixed Point in a Plane, and imagine one or more Lines passing through it, it is not difficult to grasp the following ideas—that the 'direction' of any such Line is that property of it which determines its position, now that one Point in it is already fixed—that any two such Lines form 4 angles, whose common vertex is the fixed Point—that, if one of those 4 angles be zero, the 2 Lines coincide; if not, they intersect—that, in the first case, they have the same direction, in the second, different directions—and that the difference of the directions of such Lines is measured by the angle between them.

But these ideas are of little use to us, when confronted with a given Line and a given Point outside it, and when told to imagine a new Line drawn, through the given

Point, and 'in the same direction' as the given Line, the 2 Lines having no common Point. For the directions of the Lines are no longer directly comparable. There is no use in asking "do they contain a zero-angle?" when they contain no angle whatever. An angle cannot exist without a vertex: and where is the vertex?

What idea, then, is conveyed to the mind by the phrase "these Lines have the same direction"? If they were finite Lines, and the question concerned sameness of length, the process, of grasping this idea, would be a very simple one. We could either imagine one of the 2 Lines laid upon the other, and then apply the Axiom 'magnitudes which coincide are equal': or we could imagine a movable third Line, first applied to one of the 2 Lines and ascertained to have 'the same length' with it,[Pg 69] and then carried across the intervening space and applied to the other Line; and, on finding it to have 'the same length' with that also, we should pronounce the 2 Lines to have 'the same length,' in full confidence that our movable Line had preserved its length unchanged during the journey.

Can we, then, use a similar process in grasping the idea of sameness of direction? That is, can we imagine a movable third Line, first applied to one of the 2 Lines and ascertained to have 'the same direction' with it, and then carried across the intervening space and applied to the second Line, to see if it has 'the same direction' with it also? But this process would convey to the mind no idea of 'sameness of direction,' unless we had some guarantee that the movable Line had preserved its direction unchanged during the journey. The only process, for securing this, that presents itself to my mind, is to imagine a transversal, cutting the 2 Lines, and thus bridging over the intervening space, and then to imagine the movable Line shifted along it, so as always to cut it at a constant angle.

I have thought this matter out very carefully, and I feel convinced that this is the mental process by which we grasp the idea of 'sameness of direction,' when predicated of Lines that have no common Point, and therefore cannot be said to contain a zero-angle.

Now this 'constant angle' is of course the angle at which the transversal cuts the first Line. Hence, the mental picture of a Line moving away from another, and yet maintaining 'sameness of direction' with it, is the picture of its so moving as that a certain transversal shall cut the two Lines at the same angle. And this is my answer to our first question, namely, "what idea is conveyed to the average human intellect by the phrase 'in the same direction,' when applied to non-coincident Lines?"

If this be granted, our second question, "in accepting the Axiom that 'Lines, which have the same direction, make equal angles with all transversals,' to what other assertions are we[Pg 70] committing ourselves?", must be answered "we are consciously committing ourselves to the assertion that Lines, which are equally inclined to a certain transversal, are so to any transversal."

We see, then, that, if this be so, the advocates of the Direction-Theory have not escaped the necessity of assuming, as axiomatic, the second Theorem enunciated in § 2. (See p. 63.)

I must ask the Reader's pardon for this long digression: but the fallacy, which (as I believe) lies at the root of the Direction-Theory, is a very subtle one, and has cost me a great many hours of hard thinking to un-earth it.

Yet another process has been invented—quite fascinating in its brevity and its elegance—which, though involving the same fallacy as the Direction-Theory, proves Euc. I. 32 without even mentioning the dangerous word 'Direction.'

We are told to take any Triangle ; to produce to ; to make part of , viz. , revolve, about , into the position ; then to make part of this Line, viz. , revolve, about , into the position ; and lastly to make part of this Line, viz. , revolve, about , till it lies along , of which it originally formed a part. We are then assured that it must have revolved through 4 right angles: from which it easily follows that the interior angles of the Triangle are together equal to 2 right angles.

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The disproof of this fallacy is almost as brief and elegant as the fallacy itself. We first quote the general principle that we cannot reasonably be told to make a Line fulfil two conditions, either of which is enough by itself to fix its position: e.g. given 3 Points, , , , we cannot reasonably be told to draw a Line, from , which shall pass through and : we can make it pass through , but it must then take its chance of passing through ; and vice versâ.

Now let us suppose that, while one part of , viz. , revolves into the position , another little bit of it, viz. , revolves, through an equal angle, into the position ; and that, while revolves into the position of lying along , revolves—and here comes the fallacy. You must not say "revolves, through an equal angle, into the position of lying along ," for this would be to make fulfil two conditions at once. If you say that the one condition involves the other, you are virtually asserting that the Lines , are equally inclined to —and this in consequence of having been so drawn that these same Lines are equally inclined to . That is, you are asserting § 2. (2). (See p. 63.)

One other proof—a very beautiful one, though largely dealing with Infinities and Infinitesimals—may here be mentioned, that of M. Bertrand. It rests on the principle that an infinite Sector (with a vertical angle which has a finite ratio to a right angle) is an Infinity of the same order as an infinite Plane, whereas an infinite Strip (i. e. the area contained between 2 'separational' Lines) is an Infinity of a lower order. Hence he concludes that no such Sector, however small its vertical angle, will lie wholly between 2 such 'separational' Lines, however far apart: hence, if a Line intersect one of 2 'separational' Lines, it must, if produced, intersect the other. He thus proves the Theorem numbered (8) in § 4: but his results are, of course, only partially, and not absolutely, true.

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As a rather interesting example of the ease with which an unwary explorer may tumble into a pit-fall, I may refer to a "Note on Euclid's 12th Axiom" by a Mr. W. Hanna, which will be found at p. 27 of Vol. XIII of "Mathematical Questions" reprinted from the "Educational Times." Mr. Hanna takes two Lines, situated as in Euclid's

Axiom, and drops a perpendicular, from a point on one, upon the other: from the foot of this he drops a second perpendicular back upon the first line: and so on, backwards and forwards, till the diagram slightly resembles the side of a laced-up boot. He then easily proves that these perpendiculars continually decrease in length. From this he infers that "the perpendicular will ultimately become less than any assignable line"! The fallacy is really too obvious to be worth pointing out.

Another writer in the "Educational Times" (Mr. J. Walmsley, B.A.: his article will be found at p. 103 of Vol. XVII of the Reprint) has fallen into a rather less obvious trap. He endeavours to prove that "any straight line, perpendicular to one of two parallel straight lines, will meet the other." (This, if it could be proved without assuming any disputable Axiom, would indeed be a splendid success! Mr. J. Walmsley has hardly realised, as yet, the fearful difficulty of persuading two Lines, under any conceivable circumstances, to do such a thing as "meet." Whether, at the outset of geometrical discovery, Lines were not properly introduced to each other—or whether some mischief-making Point has been insinuating that one of them went and intersected another when it was looking the other way—certain it is, that Lines will do almost anything you like to propose, rather than "meet" one another!) Mr. Walmsley assumes, as axiomatic, that when one Line lies between two others (whatever "between" may mean), those two others lie on opposite sides of it. Now let Mr. Walmsley (or any other champion of his theory) draw three Lines diverging from a Point at equal angles (of 120° each), and thus dividing the infinite Plane into three equal Sectors. In each of these Sectors let him draw a branch of a Hyperbola, having the sides of the Sector as its Asymptotes. Now, each of these Hyperbolæ lies (in a way) "between" the other two: and yet no two can be said to lie on opposite sides of the other one! "But," says Mr. Walmsley (or the champion aforesaid) "these are Curves, not straight Lines!" Most true: but how are you to know that straight Lines will not behave just like Hyperbolæ, if only they are put far enough apart? Produce those three radiating Lines, that we began with, until each is a million miles long (paper, pen, and ink, provided regardless of expense): then, across their extremities, draw three Lines perpendicular to them. Why shouldn't these three Lines (of course shunning a "meeting," as all Lines do) perpetually face inwards, so that no one of them ever commits the discourtesy of turning its back upon either of the other two? "It cannot be," say Messrs. Walmsley and Co.: "these Lines must intersect, if produced far enough." What? In consequence of their relative situation? That relative situation being, for each pair of them, that they make with a certain transversal two interior angles together less than two right angles? In assuming this, I very much fear that Messrs. Walmsley and Co. are performing the not-wholly-unprecedented feat of assuming Euclid's 12th Axiom!

§ 6.

[Pg 73]

The Outlook.

In conclusion let me address myself to the young and eager explorer who, Alpine-staff in hand, and duly furnished with all the necessaries for his perilous

quest—pick-axe, theodolite, paper-collars, Brown's Sticking-Plaster, Jones's Cough-Pills, and Robinson's Insect-Powder, "the only known remedy for Phlebitis"—is preparing to sally forth, to do or die!

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To him let me address myself, as being, perchance, an older and a more experienced traveller—one who has wandered much, and pondered long, and who can best describe himself in the words of a lady well-known in the literary world (I am glad to have this opportunity of recording them, as they have never been printed. They were written "for music," for which purpose, I imagine, the amount of sense required is not excessive).

"I have wandered,

I have pondered,

I have squandered

Many a boon:

In the sadness,

In the gladness,

In the madness

Of the moon.

"Seek thy pillow

By the billow,

Where the willow

Doth not weep:

Few will wonder

Who lies under,

Hearing thunder,

Fast asleep!"

Poetry like this speaks for itself: vain were it to hope that any poor words of mine could serve to illuminate, or even elucidate, its almost ethereal beauty!

To what point of the compass, then, should this young and eager explorer be advised to direct his steps?

I think his best chance—and that only a slender one—is to find some elementary proof for my Axiom, or for one of the many Theorems which will serve the same purpose, a few of which I will enumerate. In the first place, any Polygon will do, and any ratio, between it and the out-lying Segment, so long as[Pg 75] it is a finite ratio. What I want it for is to prove that there is some isosceles Triangle, with a definite vertical angle (i. e. some named fraction of a right angle), whose base is less than one

of its sides. And that, again, is wanted in order to prove it possible to draw, on a given base, an isosceles Triangle, whose base-angles shall have some nameable value. And that, again, is wanted in order to prove that there is some finite minimum value for the sum of the angles of a Triangle. And that, again, is wanted in order to prove Prop. 3, at p. 19. So, if any one of these propositions could be either assumed as an Axiom or proved as a Theorem, it would suffice for the proof of Euc. I. 32 and Co.

If, for example, you will grant me, as an Axiom, that there is some finite minimum value for the sum of the angles of a Triangle, no matter how small you make it, all is easy at once: grant me, for instance, that no Triangle can possibly have the sum of its angles less than a millionth of a right angle, and I am happy!

Finally, I am inclined to believe that, if ever Euc. I. 32 is proved without a new Axiom, it will be by some new and ampler definition of the Right Line—some definition which shall connote that peculiar and mysterious property, which it must somehow possess, which causes Euc. I. 32 to be true. Try that track, my gentle Reader! It is not much trodden as yet. And may success attend your search!

THE END.

[TURN OVER.

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